



NADRA | STATE BANK BUILDING, SECTOR G-5/2 ISLAMABAD

Biometric Verification Service

SPECIFICATION DOCUMENT

Version: 1.1

Contents

1	Intended Audience.....	3
2	Introduction	3
3	Assumptions/Requirements	3
4	Constraints	4
5	Technical Interface Specifications.....	4
5.1	Operations	4
5.2	Biometric Verification	4
5.2.1	Fingerprint Verification	5
5.2.2	Last Verification Result	5
6	Biometric Standards.....	6
6.1	Fingerprint Standards	6
7	Annexure – A.....	7
7.1	Verification.....	7
7.1.1	Fingerprint Verification	7
7.1.2	Last Verification Result	9
7.2	Generic Codes description	10

1 INTENDED AUDIENCE

This document is intended for

- The Developers who can review project's capabilities and more easily understand where their efforts should be targeted to improve or add more features to it (design and code the application and sets the guidelines for future development).
- The End users of Biometric Verification Service BioVeriSys who wish to read about what this project can do. It also addresses the end user's technical team regarding the details of interfaces necessary to use the NADRA's Biometric Verification Service BioVeriSys.

2 INTRODUCTION

NADRA is providing service providers with a Biometric Verification Service named as "BioVeriSys". BioVeriSys aims to bring efficient and instantaneous verification. The architecture of Service has been intelligently engineered to offer an enterprise, interoperable, scalable and flexible service for the market. The service is completely finger print device agnostic and uses multiple fingerprint format.

3 ASSUMPTIONS/REQUIREMENTS

- This service will be a point to point connection between NADRA and service providers.
- Interfacing with NADRA BioVeriSys is through XML's as more specially specified through XSD
- Biometric verification shall be performed at NADRA's server side.
- Service Provider shall be issued a unique Franchisee ID and username password for data exchange and authentication.
- Service providers shall be responsible for providing data necessary for performing biometric verification at server side.
- Service provider will provide finger print data on following allowance template:-
 - ANSI
 - ISO_19794_2
 - SAGEM_PKMAT
 - SAGEM_PKCOMPV2
 - SAGEM_CFV
 - RAW_IMAGE
 - WSQ

4 CONSTRAINTS

Following are the key constraints of the BioVeriSys web service:

- If citizen status is not verified in National Citizen Database e.g. citizen marked as duplicate, suspect, alien, fraud, then verification will not be carried out and the service will respond with an appropriate message to service providers, directing the customer to visit NADRA center.
- Once the session has been created, service providers can verify by using any of the available four fingers. Each finger can have only two verification attempts. Hence service asks for another finger for verification in case a finger had two failed verification attempts.
- In one session service provider can verify one finger 2 times out of 4 available fingers (total 8 attempts)
- Once verification is successful session will be expired.
- Each session is a billable entry

5 TECHNICAL INTERFACE SPECIFICATIONS

The data exchange interfaces between NADRA Biometric Verification Service and Service Providers is through XML web services. The accompanying XSD's are also shared along with this specification document.

5.1 Operations

The following are methods that are available for performing biometric verification transaction. Each method is initiated with a request to NADRA Biometric Verification Service that carries all the data corresponding to the nature of request in XML format. Biometric Verification Service responds back with a response XML carrying the responses corresponding to its accompanying request.

NADRA shall also share XML definition in form of XSD for request/response interfaces. The Service provider request is composed of Franchisee ID and an XML containing the method Signature and all the parameters of that request. The response includes XML which contains response data specified in XSD.

5.2 Biometric Verification

In first step Service verifies the citizen from National Citizen Database and replies with the appropriate message. If the person is a valid citizen and the data provided for verification is according to the specified XML format, the web service performs 1:1 fingerprint verification. If the provided fingerprint is verified, the Service will return citizen data along with the verification results. And if the fingerprint verification is not successful, the service will return 4 available finger indexes.

NOTE:

- *Service will generate unique session id.*
 - *This session id is billable entry and will remain valid for next 10 minutes*
 - *The 8 attempts for fingerprints verification can be requested.*
 - *Each fingerprint can be submitted only twice for verification.*
 - *If the verification is successful, the session will expire automatically.*
- The operation takes input as mentioned in *Annexure A: 7.1.1. Fingerprint Verification*

PROCESS

- **Verify Citizen Status from National Citizen Database:** If person is not verified, operation will return appropriate message along with session id. Fingerprint verification will not be carried out.
- **Verify Fingerprints Availability:** If fingerprints are not available, person should be advised to visit NADRA NSRC for modification of CNIC.
- **Verify Fingerprints:**
 - If verification is successful operation will return session id and session will be closed for given citizen no.
 - If verification is unsuccessful operation will return session id, and 4x available finger indexes. Fingerprint verification can be re-called for any of 4x available fingers with same session id, same citizen number & same transaction ID which was used to generate session ID. Maximum number of finger print verifications attempts for per session will be 8.

NOTE: Session id is a billable entry

OUTPUT

- The operation produces a response as mentioned in *Annexure A: 7.1.1. Fingerprint Verification*

5.2.2 Last Verification Result

DESCRIPTION

The Service provider can request for the last verification result from BioVeriSys before re-sending the request to Finger Verification.

PRE-REQUISITES

This operation is called by the Service provider whereby the instantaneous verification result from the Finger Verification is not received in the stipulated time.

INPUT

- The operation takes input as mentioned in *Annexure A: 7.1.2. Last Verification result*

PROCESS

- **Verify Session and last verification results:** This function checks whether session exists and any verification was done against citizen no.
- **Verification result:** If verification exists service will provide appropriate message against the verification.

OUTPUT

- The operation produces a response as mentioned in *Annexure A: 7.1.2. Last Verification result*

6 BIOMETRIC STANDARDS

6.1 Fingerprint Standards

- Finger prints should be acquired at 500 dpi before conversion to any template format
- Maximum information of fingerprint should be acquired
- Fingerprint should be placed flat with maximum contact between scanner lens and skin
- Image lens should be clean and without any moisture for better biometric acquisition
- Fingerprint template should be one of following
 - ANSI
 - ISO_19794_2
 - SAGEM_PKMAT
 - SAGEM_PKCOMPV2
 - SAGEM_CFV
 - RAW_IMAGE
 - WSQ
- Finger Index definition

Finger Index	Description
1	Right Thumb
2	Right Index Finger
3	Right Middle Finger
4	Right Ring Finger
5	Right Little Finger
6	Left Thumb
7	Left Index Finger
8	Left Middle Finger
9	Left Ring Finger
10	Left Little Finger

7 ANNEXURE – A

7.1 Verification

7.1.1 Fingerprint Verification

7.1.1.1 Request Interface

Field Name	Field type	Length	Constraints (if any)	Description	Example
Franchisee ID	String	4	Unique	The unique Franchisee ID issued to Service provider by NADRA	1234
Username	String	Max 12 characters	Mandatory	Username of Service provider issues by NADRA for BioVeriSys	iuser
Password	String	Max 15 characters	Mandatory	Password set by the Service provider for BioVeriSys	ipassword
Session ID	String	19	Non Mandatory	A unique 19 digit number	1000009010000090
Service Provider Transaction ID	String	19	Unique Numeric	This is the unique transaction id that Service provider provided for verification	6407814020617511501
Citizen Number	String	13	Mandatory	This is the citizen number for which session id is being requested	6110119876547
Contact Number	String	16	Non Mandatory	Contact no of citizen e.g mobile no.	0308-5126899
Mobile Number	String	16	Mandatory	Mobile no from which verification is performed.	+923038660795
Longitude	String		Mandatory	Longitude of device in decimal degree format from which verification is performed	65.074920
Latitude	String		Mandatory	Latitude of device in decimal degree format from which verification is performed	33.442988
IMEI	String	Min 14 Max 16	Mandatory	International Mobile Equipment Identity (IMEI) number of device from which verification is performed	369725661479536
Finger Index	Integer		Mandatory	Finger index for which verification is attempted. See Section 6.1	1
Finger Template	Base 64 string		Mandatory	This is the finger template corresponding to the finger print index	

Template Type	String		Mandatory	Value of Template type from Template Type data type exposed by Service	TemplateType.PKMAT
Area Name	String	32	Mandatory	Name of area from where verification is being requested.	• khyber-pakhtunkhwa • fata • punjab • sindh • baluchistan • islamabad • gilgit-baltistan • azad Kashmir

7.1.1.2 Response Interface

Field Name	Field type	Length	Constraints (if any)	Description	Example
Code	String	3	Mandatory	Code response to request	100
Message	String	256	Mandatory	Message explaining the response code	Successful
Session ID	String	19	Mandatory	Unique session id for using BioVeriSys. This session ID will be used for the billing to the service providers	4001100000103719929
Citizen Number	String	13	Mandatory	Contains Primary/Sender citizen number	6110119876547
Finger Index	String[]		Non Mandatory Where verification result is not successful	This is the list of available finger indexes	1,2,4,6
Response Parameter(s)	Please refer to XSD folder for agreed Response Parameter(s)				

7.1.1.3 Code description

Sr. No.	Code	Description
1	100	successful
3	110	citizen number is not verified
4	111	finger not found in citizen database
5	112	error generating session id
7	114	invalid verification reference number
8	115	Invalid service provide transaction id
9	118	finger verification has been exhausted for current finger
10	119	verification limit for current citizen number has been exhausted
11	120	invalid input finger template
12	121	invalid finger index
13	122	fingerprint does not matched
14	123	invalid finger template type
16	125	contact number is not valid
17	175	transaction id already exist

18	185	Invalid area name
19	186	invalid longitude
20	187	invalid longitude
21	188	invalid imei
22	189	this service can only be accessed from Pakistan
23	190	mobile number is not valid

7.1.2 Last Verification Result

7.1.2.1 Request Interface:

Field Name	Field type	Length	Constraints (if any)	Description	Example
Franchisee ID	String	4	Unique	The unique Franchisee ID issued to Service provider by NADRA	1234
Username	String	Max 12 characters	Mandatory	Username of Service provider issued by NADRA for BioVeriSys	iuser
Password	String	Max 15 characters	Mandatory	Password set by the Service provider for BioVeriSys	ipassword
Service Provider Transaction ID	String	19	Unique per session Numeric	This is the unique transaction id that Service provider provided for verification	6407814020617511501
Citizen Number	String	13	Mandatory	This is the primary/sender citizen number for which the session id is issued	6110119876547

7.1.2.2 Response Interface:

Field Name	Field type	Length	Constraints (if any)	Description	Example
Code	String	3	Mandatory	Code response to request	100
Message	String	256	Mandatory	Message contains last verification result	Successful
Session ID	String	19	Mandatory	A unique 19 digit number issued	1000009010000090
Citizen Number	String	13	Mandatory	This is the primary/sender citizen number for which the session id is issued	6110119876547

7.1.2.3 Code Description

Sr. No.	Code	Description
1	100	successful
2	150	invalid citizen number
3	151	no request found against transaction id
4	152	Last verification was not successful

7.2 Generic Codes description

Sr. No.	Code	Description
1	102	invalid xml
2	103	invalid user name/password
3	104	Invalid session id
4	201	session has been expired
5	202	verification was successful therefore session has been expired
6	105	user do not has access to this functionality
7	106	Invalid franchisee id
8	107	invalid citizen number
9	108	Citizen verification Service is down. Please try again later.
10	109	Exception: system has encountered an unexpected error. Administrator has been informed, please try again later.